

Servers for AI uses huge amount of energy. How much carbon footprint does all these servers have compared to cars?

Posted by u/Nervous_Type_9175 • 0 points • 9 comments

For eg: can one say that all AI servers consume energy per equivalent to 50,000 cars per year? Has anyone come across some studies in this area?

Comments

u/SpeedyHAM79 • 1 points • 2025-10-18 06:54 UTC

Entirely depends on the energy supply source for the grid that the data center is located and the size of the data centers. Some new data centers are being designed around a continuous 1 Gw electrical load.

u/Enough_Island4615 • 1 points • 2025-10-17 22:17 UTC

Worldwide, AI uses just under 2% of the energy that road transport uses.

u/bangbangracer • 1 points • 2025-10-17 13:34 UTC

It's a small percentage compared to things like cars, but this is why pure numbers aren't really a thing and you need to add context.

In economics, there's a concept called "Utils" and that's a way to talk about the utility of something. A t-shirt will have the same utils as any other t-shirt regardless of price. I bring this up because cars have oodles of utils simply because the US transit system is built around them. AI isn't like that. The utils are low with AI. It can't build a house. It can design one, but we also have existing house designs. So when you see something like "training an AI model has the same footprint as 5 cars over their lifetime", that's a big deal. A car has moved people literal hundreds of thousands of miles over its estimated 10 year lifespan. That AI model hasn't even done anything yet.

u/Nervous_Type_9175 • 0 points • 2025-10-17 14:55 UTC

True. But each AI prompt is consuming energy irrespective of the utils.

u/KronusIV • 1 points • 2025-10-17 12:50 UTC

Depends on where the power is coming from. Microsoft is working on restarting Three Mile Island to power their data center, that's a very low carbon footprint.

u/Nervous_Type_9175 • 1 points • 2025-10-17 14:54 UTC

Ja, but the power generation will use fossil, or renewable or nuclear. How much energy is used?

u/KronusIV • 2 points • 2025-10-17 16:04 UTC

Three Mile Island is a nuclear power plant.

u/Nervous_Type_9175 • 1 points • 2025-10-17 16:10 UTC

Ok. So how much electricity does AI use compared to EV? How many EVs will drive per month compared to the AI power consumption?

u/thatOneBl1p • 1 points • 2025-10-17 12:41 UTC

Here's an article— <https://news.mit.edu/2025/responding-to-generative-ai-climate-impact-0930> it's not strictly about cars but there are some helpful comparisons, one of which is driving.

Here's another one with more numbers—

<https://www.technologyreview.com/2019/06/06/239031/training-a-single-ai-model-can-emit-as-much-carbon-as-five-cars-in-their-lifetimes/amp/>